

Khin Eaindray Htun

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Education

King Mongkut's University of Technology Thonburi <i>Master of Engineering in Electrical and Information Engineering</i>	Bangkok, Thailand Aug. 2025 – May 2027
King Mongkut's University of Technology Thonburi <i>Bachelor of Science in Computer Science, First Class Honors</i>	Bangkok, Thailand Aug. 2022 – May 2025

Professional Experience

Deep Learning Lab, KMUTT <i>Research Member</i>	Bangkok, Thailand Dec. 2023 – Present
- Developed a high-resolution crack detection system using YOLO with attention modules, achieving 87% accuracy on private datasets for infrastructure inspection. - Built and evaluated an ensemble YOLOv8 solution for real-time corrosion detection from high-resolution drone imagery, achieving 75% IoU on Skyller's private dataset. - Conducted model experimentation and performance evaluation for industry-sponsored R&D projects, applying deep learning to real-world inspection problems.	
Specific Impulse Co., Ltd <i>Software Engineer (AI / Computer Vision)</i>	Bangkok, Thailand Jun. 2025 – Feb. 2026
- Integrated DJI Dock and aircraft telemetry into a frontend monitoring dashboard using DJI Cloud APIs and JSON-based data for near-real-time operational visibility. - Developed a proof-of-concept integration for newer DJI platform/API versions, adapting backend interfaces and data flows after support for the existing integration was discontinued. - Designed and deployed real-time AI and edge-processing systems, including a 10 FPS image-processing pipeline and Orange Pi CM5 deployment, while evaluating LoRa communication latency, packet loss, and reliability.	
Agoda <i>Data Engineer Intern</i>	Bangkok, Thailand Jun. 2024 – Oct. 2024
- Optimized SQL-based Kubeflow data-processing workflows, reducing batch processing time from 20 minutes to 2 seconds and improving efficiency by 99.83%. - Developed an automated housekeeping tool for Kubeflow Pipelines to manage outdated data across S3 and VastFS-backed storage. - Collaborated with engineering teams to integrate pipeline components into Kubernetes-based production workflows, improving scalability and reliability.	

Selected Projects

Autonomous Drone Simulation & Real-Time Sensor Data Pipeline <i>Isaac Sim, Python, Pegasus SDK, HDF5, OpenCV</i>	
- Built a 300×300 m NVIDIA Isaac Sim environment with five autonomous UAVs and four RGB-D cameras for multi-view tracking and simulation-based data generation. - Developed a Python pipeline using the Pegasus API to control UAV movement, capture synchronized RGB/depth data at 10 FPS, and store ML-ready outputs in HDF5.	
Flu Shot Learning – End-to-End ML Classification <i>Python, Scikit-learn, XGBoost, CatBoost, LightGBM</i>	
- Applied the CRISP-DM lifecycle for data understanding, preprocessing, model development, and evaluation to predict H1N1 and seasonal flu vaccine uptake. - Compared multiple classification models after KNN and mode-based imputation, achieving a DrivenData competition score of 0.8632 and ranking 121st.	
Ezria – RAG-Based AI Study Assistant <i>Python, Flask, AWS, Docker, Generative AI, RAG</i>	
- Developed a RAG-based application that retrieves information from uploaded documents and generates contextual explanations and question-answering responses. - Designed and evaluated the LLM retrieval pipeline, improving response relevance through preprocessing and prompt optimization, with deployment using Flask, AWS, and Docker.	

Technical Skills

Programming & Data: Python, SQL, R, Pandas, NumPy
Databases: PostgreSQL, MySQL
Machine Learning & AI: Scikit-learn, PyTorch, TensorFlow, XGBoost, LightGBM, Feature Engineering, Generative AI, Retrieval-Augmented Generation (RAG)
Data Engineering & MLOps: Kubeflow, Kubernetes, Docker, AWS S3, EC2, Lambda, Data Pipelines, CI/CD, Model Monitoring
Backend & Integration: Flask, REST APIs, JSON, API Integration, Real-Time Data Processing
Analytics & Visualization: EDA, Data Cleaning, Statistical Analysis, Power BI, Matplotlib, Plotly
Simulation & Industrial Systems: NVIDIA Isaac Sim, Pegasus SDK, OpenCV, HDF5, RGB-D Data, IoT Telemetry, Edge Computing, LoRa